

Continuing with Jupyter Notebook

📺 How to Create New Environment in Jupyter Notebook (2024) - Jupyter Notebook Virtual E...

Video goes over the process of creating first the kernel for a virtual environment: establishing a name and a dedicated directory for the kernel, which will then be used for future files utilising the kernel for access to the virtual environment.

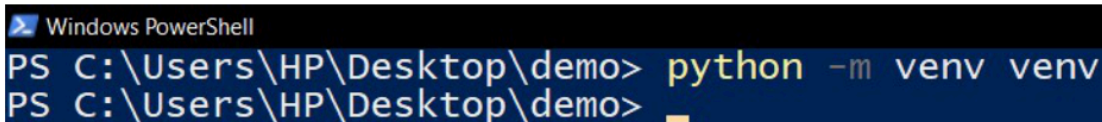
<https://www.geeksforgeeks.org/python/using-jupyter-notebook-in-virtual-environment/>

Geeksforgeeks article shows the same basic process

Step 1: Create a virtual environment

Open the directory where you want to create your project. open cmd/powershell and navigate to the same directory and run the following commands to create a virtual environment.

```
python -m venv venv
```



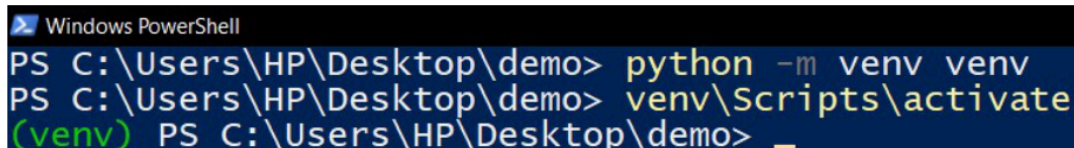
```
Windows PowerShell
PS C:\Users\HP\Desktop\demo> python -m venv venv
PS C:\Users\HP\Desktop\demo> _
```

*python -m venv creates the virtual environment, then name for the ve follows

Step 2: Activate the virtual environment

Now as we have our virtual environment let's activate it.

```
venv\Scripts\activate
```



```
Windows PowerShell
PS C:\Users\HP\Desktop\demo> python -m venv venv
PS C:\Users\HP\Desktop\demo> venv\Scripts\activate
(venv) PS C:\Users\HP\Desktop\demo> _
```

```
(NEW-ENV) C:\Users\visha\OneDrive\Desktop\NewKernel>pip install ipykernel
```

*pulled from the video, stated to be installed so it may function as a kernel in jupyter notebook

Step 3: Install jupyter kernel for the virtual environment using the following command:

Running the following command will create a kernel that can be used to run jupyter notebook commands inside the virtual environment.

```
ipython kernel install --user --name=venv
```

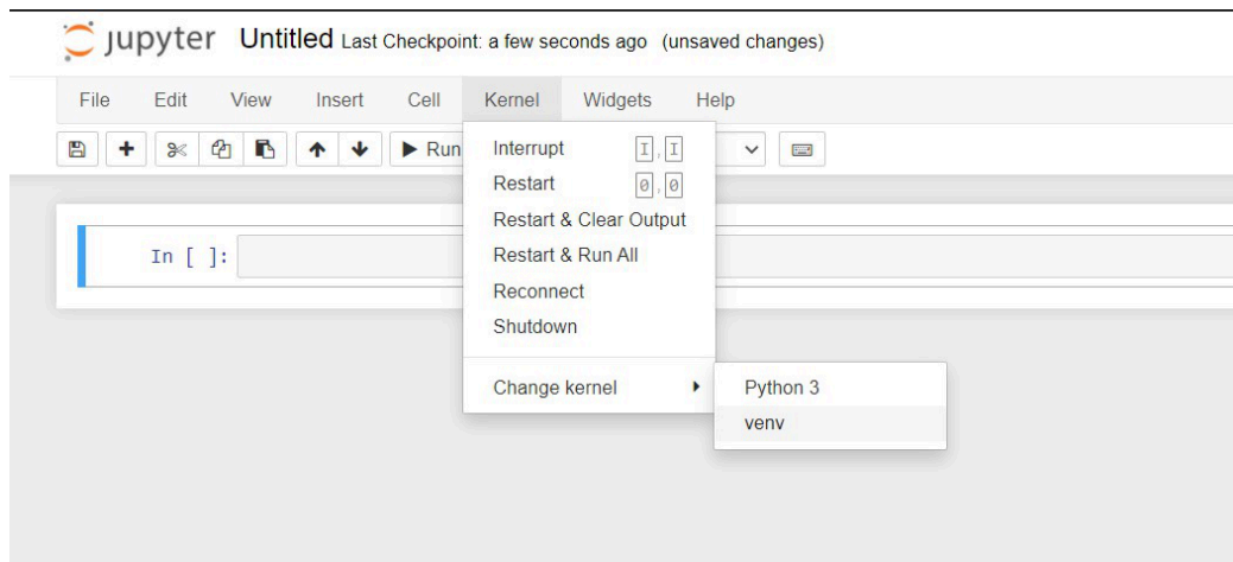
```
(venv) PS C:\Users\HP\Desktop\demo> ipython kernel install --user --name=venv
Installed kernelspec venv in C:\Users\HP\AppData\Roaming\jupyter\kernels\venv
(venv) PS C:\Users\HP\Desktop\demo>
```

```
(NEW-ENV) C:\Users\visha\OneDrive\Desktop\NewKernel>python -m ipykernel install --user --name MyNewEnv
```

*using examples from both article and video

Step 4: Select the installed kernel when you want to use jupyter notebook in this virtual environment

Let's now check if our kernel is created. Just run "jupyter notebook" command in the command prompt or Powershell and the jupyter environment will open up. Click on the kernel and click change kernel you will be able to see the kernel you just created.



** Video ends by saying any additional downloads and resources can be added into the virtual environment through the terminal and would show in the kernel, still researching that

3) Install once

```
1 pip install -r requirements.txt
2 pip install ipykernel
3 python -m ipykernel install --user --name my_suite_venv --display-name "Python (.venv)"
4 ``
```

✓ Pros: simple, standard

⚠ Cons: harder when you need non-Python system libraries

Installing the necessary libraries/ dependencies seems to be done at the same time as installing the ipykernel, gonna find if that order matters, or if installations can be done regardless of order

Resources:

Using Jupyter Notebook with Google Colab

Using 'Apptainer'

Found article:

[Custom Jupyter Notebooks using apptainer | Minnesota Supercomputing Institute](#)

Info of note: goes into explanation on creating an image to use with apptainer to create a container (like a secure environment) with set dependencies and all necessary metadata

Installation with CMake:

[How-to Guide: Set Up for CMake-based builds | google-cloud-cpp](#)

[CMake Tutorial for Absolute Beginners - From GCC to CMake including Make and Ninja](#)

Official Apptainer Website

Web- [Get Started | Apptainer - Portable, Reproducible Containers](#)

Github - <https://github.com/apptainer/apptainer?tab=readme-ov-file>

Quick start documentation - [Quick Start — Apptainer User Guide main documentation](#)